

ICS 4111: Embedded Systems & IoT

Practical Exercise 2: Component Tables

Part B

Question 1a

No.	Component	Symbol	Value	Quantity	Function
1	Power Supply	VCC	125 V	1	Provides electrical energy to the circuit
2	Resistor	R1	20 Ω	1	Opposes current flow
3	Resistor	R2	30 Ω	1	Opposes current flow
4	Resistor	R3	50 Ω	1	Opposes current flow
5	Ground	GND	0 V	1	Completes the circuit and discharges excess current
6	Connecting Wires	—	—	4	Connections between components

Question 1b

No.	Component	Symbol	Value	Quantity	Function
1	Power Supply	VCC	125 V	1	Provides electrical energy to the circuit
2	Resistor	R1	20 Ω	1	Opposes current flow
3	Resistor	R2	100 Ω	1	Opposes current flow
4	Resistor	R3	50 Ω	1	Opposes current flow
5	Ground	GND	0 V	1	Completes the circuit and discharges excess current
6	Connecting Wires	—	—	6	Connections between components

Question 2

No.	Component	Symbol	Value	Quantity	Function
1	Arduino Nano	U1	—	1	Microcontroller that reads sensor data
2	DHT22 Sensor	U3	—	1	Measures temperature and humidity
3	Connecting Wires	—	—	3	Connections between components